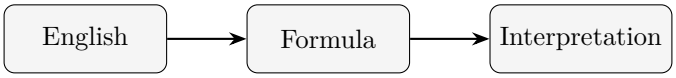
DISCRETE STRUCTURES

CMSC250

Chapter 7: Interpreting Mathematical Statements

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Domains, translations, interpretations, and quantified reasoning



Notation. We use $\sim$ for negation; $\neg$ is an equivalent alternative. Quantifiers are always interpreted relative to a specified domain.

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1 From formulas to meaning

A formula such as $\forall x\, P(x)$ is not automatically true or false until we know what objects x ranges over and what P means. An **interpretation** supplies that missing information. In this chapter we learn to translate carefully, construct interpretations, and decide whether a statement is true, false, or logically valid.

For instance, $\forall x\,(x^2 \geq x)$ is true over $\mathbb{N} = \{0, 1, 2, \ldots\}$ but false over $\mathbb{R}$: the real number $x = \frac{1}{2}$ is a counterexample. The symbols did not change; the domain did.

Key idea. Always separate the *syntax* (the written formula) from its *semantics* (what it means under a domain and interpretation).

2 The three pieces of an interpretation

An interpretation must specify (1) a nonambiguous **domain** D; (2) the meanings of predicates such as $P(x)$ and $R(x, y)$; and (3) where relevant, the objects named by constants. A predicate with two arguments is interpreted as a relation on ordered pairs, so $R(a, b)$ and $R(b, a)$ need not agree.

Worked example: the same syntax can have opposite truth values

Let $D = \{a, b\}$. In Interpretation I, set $P(a) = 1$ and $P(b) = 1$. Then $\forall x\, P(x)$ is true. In Interpretation II, set $P(a) = 1$ and $P(b) = 0$. Then $\forall x\, P(x)$ is false, but $\exists x\, P(x)$ remains true. The latter has witness a; the former has counterexample b.

3 Domains and restricted quantifiers

When a problem gives all people as the domain, let $S(x)$ mean "x is a student" and $H(x)$ mean "x completed the homework." Then

$$\text{Every student completed the homework} \quad \longleftrightarrow \quad \forall x\,(S(x) \to H(x)).$$

The implication matters. A person who is not a student should not make the sentence false. On the other hand,

$$\text{Some student completed the homework} \quad \longleftrightarrow \quad \exists x\,(S(x) \wedge H(x)).$$

The conjunction matters because a witness must be *both* a student and someone who completed the homework.

Key idea. For restricted universal claims, usually use an implication. For restricted existential claims, usually use a conjunction.

Worked example: why the tempting alternatives fail

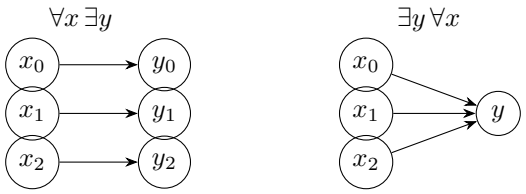
The formula $\forall x\,(S(x) \wedge H(x))$ says *everyone is a student and everyone completed the homework.* That is stronger than "every student completed it." The formula $\exists x\,(S(x) \to H(x))$ can be true merely because a nonstudent exists: for such a person $S(x)$ is false, making the implication true regardless of $H(x)$.

4 Nested quantifiers and dependencies

Consider the following two statements over integers:

$$A: \ \forall x \, \exists y \, (y > x), \qquad B: \ \exists y \, \forall x \, (y > x).$$

A is true: given an arbitrary x, choose $y = x + 1$. The witness y may *depend on* x. B is false: no fixed integer exceeds every integer, because for any proposed y we can choose $x = y + 1$.



The left diagram allows a different witness for each x. The right diagram requires one common witness. This is the central meaning of quantifier order.

5 Translating English precisely

Before writing symbols, identify the domain, the predicates, the quantifier order, and whether a claim is conditional or conjunctive.

Worked example: everyone knows someone

Let $K(x, y)$ mean "x knows y" over the domain of people. "Everyone knows someone" becomes

$$\forall x \, \exists y \, K(x, y).$$

"There is someone whom everyone knows" becomes

$$\exists y \, \forall x \, K(x, y).$$

These are not equivalent. A community where each person knows only themselves satisfies the first, but not the second if the community has at least two people.

Worked example: every student solved at least one problem

With students as the domain of x, problems as the domain of y, and $C(x, y)$ meaning "student x solved problem y," write

$$\forall x \, \exists y \, C(x, y).$$

The formula $\exists x \, \exists y \, C(x, y)$ merely says *some* student solved *some* problem. Accidentally changing the initial $\forall$ to $\exists$ changes the actual mathematical claim.

6 At least one, at most one, and exactly one

Suppose $P(x)$ means "x passed."

$$\begin{aligned} \text{At least one:} &\quad \exists x\, P(x). \\ \text{At most one:} &\quad \forall x\, \forall y\, ((P(x) \wedge P(y)) \to x = y). \\ \text{Exactly one:} &\quad \exists x\, (P(x) \wedge \forall y\, (P(y) \to y = x)). \end{aligned}$$

"At most one" permits *zero* passing students. "Exactly one" combines existence and uniqueness.

Worked example: at least two distinct witnesses

"At least two people passed" is

$$\exists x\, \exists y\, (P(x) \wedge P(y) \wedge x \neq y).$$

Without $x \neq y$, the same person could be used twice, so the formula would only guarantee one witness.

7 Negation of nested quantified statements

To negate a quantified statement, move the negation inward by switching each quantifier and negating the final predicate:

$$\sim \forall x\, P(x) \;\equiv\; \exists x \sim P(x), \qquad \sim \exists x\, P(x) \;\equiv\; \forall x \sim P(x).$$

Worked example: negating a two-quantifier sentence

Negate "Every student solved some problem":

$$\begin{aligned} &\sim (\forall x\, \exists y\, C(x, y)) \\ \equiv\; &\exists x \sim (\exists y\, C(x, y)) \\ \equiv\; &\exists x\, \forall y \sim C(x, y). \end{aligned}$$

In words: "There is a student who solved no problems." This is *not* the same as "Every student solved no problems."

Worked example: negating a relational inequality

Over the real numbers, negate $\forall x\, \exists y\, (x < y)$:

$$\sim (\forall x\, \exists y\, (x < y)) \equiv \exists x\, \forall y\, (x \geq y).$$

Notice that the final inequality is $\geq$, not $>$: the negation of $x < y$ is $x \geq y$.

8 Constructing counterinterpretations

To prove that two formulas are *not* logically equivalent, find one interpretation where they have different truth values. You do not need to check every possible interpretation.

Worked example: distributing $\forall$ over OR fails

Compare

$$A = \forall x\,(P(x) \lor Q(x)), \qquad B = (\forall x\,P(x)) \lor (\forall x\,Q(x)).$$

Let $D = \{a, b\}$ with $P(a) = 1$, $P(b) = 0$, $Q(a) = 0$, and $Q(b) = 1$.

Object	P	Q	$P \lor Q$
a	1	0	1
b	0	1	1

Every object satisfies $P \lor Q$, so A is true. But $\forall x\,P(x)$ and $\forall x\,Q(x)$ are both false, so B is false. Thus $A \not\equiv B$.

Worked example: distributing $\exists$ over AND fails

Compare $\exists x\,(P(x) \land Q(x))$ with $(\exists x\,P(x)) \land (\exists x\,Q(x))$. In the same interpretation, P has witness a and Q has witness b, so the second formula is true. No object satisfies both, so the first is false.

Key idea. For a counterinterpretation, explicitly specify the domain and predicate values, then evaluate both formulas. An unexplained assertion that they differ is not a complete argument.

9 Validity versus truth in one interpretation

A formula is **satisfiable** if it is true in at least one interpretation. It is **valid** if it is true in every permitted interpretation. It is **unsatisfiable** if it is false in every interpretation. These are different claims.

For instance, $\exists x\,P(x)$ is satisfiable (take a one-element domain with P true) but not valid (take a nonempty domain with P false everywhere). In contrast, assuming classical logic and a fixed domain, $\forall x\,(P(x) \lor \sim P(x))$ is valid: for each object, $P(x)$ is either true or false.

10 Empty domains and vacuous truth

If the domain is empty, $\forall x\,P(x)$ is true because there is no counterexample, while $\exists x\,P(x)$ is false because there is no witness. Thus

$$\forall x \in \varnothing\,P(x) \text{ is true}, \qquad \exists x \in \varnothing\,P(x) \text{ is false}.$$

This convention makes the negation laws work even for empty domains. Some settings assume a nonempty domain by default, so read the problem statement carefully.

11 Quantified reasoning: useful and invalid moves

If $\forall x\,P(x)$ holds and a belongs to the domain, then $P(a)$ follows by **universal instantiation**. If $P(a)$ holds for a specific a, then $\exists x\,P(x)$ follows by **existential generalization**. The reverse directions require care.

Worked example: a valid inference

Premises: $\forall x\,(S(x) \to H(x))$ and $S(a)$. By universal instantiation, $S(a) \to H(a)$. By modus ponens, $H(a)$. This is a valid conclusion about the particular person a.

Worked example: an invalid inference

From $\exists x\,P(x)$ we cannot conclude $P(a)$ for a preselected object a. The existential witness may be someone else. Likewise, knowing $P(a)$ for one specific person does not prove $\forall x\,P(x)$.

12 Common mistakes and grading pitfalls

1. **Quantifier substitution:** $\forall x$ and $\exists x$ are not interchangeable. One wrong quantifier can reverse the claim.
2. **Wrong connective for restrictions:** use $S \to P$ for "every student" and $S \wedge P$ for "some student."
3. **Changing the order:** $\forall x \exists y$ permits a dependent witness; $\exists y \forall x$ requires one witness for all.
4. **Missing distinctness:** "two people" requires $x \neq y$.
5. **Incomplete negation:** flip each quantifier and negate the innermost proposition, including $<$ versus $\geq$.
6. **Unspecified interpretation:** a counterexample must provide the domain and the relevant predicate values.
7. **Vague conclusion:** explicitly state which formula is true and which is false in the counterinterpretation.
8. **Confusing truth and validity:** true in one interpretation is not the same as true in all interpretations.

13 Practice problems

Solve each before checking the solutions. Unless otherwise stated, domains are nonempty.

1. With all people as the domain, $S(x)$ = "x is a student" and $R(x)$ = "x reads." Translate "Every student reads."

2. Using the same predicates, translate "Some student does not read."

3. With students and assignments as respective domains, $F(x, y)$ = "student x finished assignment y." Translate "Every student finished at least one assignment."

4. Translate "There is an assignment that every student finished." Explain how it differs from Problem 3.

5. Negate $\forall x\,\exists y\,F(x, y)$, with negation pushed inward.

6. Over $\mathbb{Z}$, decide whether $\forall x \exists y\,(x + y = 0)$ is true. Provide a witness rule.

7. Over $\mathbb{Z}$, decide whether $\exists y \forall x\,(x + y = 0)$ is true. Justify.

8. Write "exactly one person is late" using $L(x)$ over people.

9. Write "at least two distinct people are late."

10. Write "at most one person is late." Does this imply anyone is late?

11. Find a two-element interpretation where $\forall x(P(x) \vee Q(x))$ is true but $(\forall x P(x)) \vee (\forall x Q(x))$ is false.

12. Find an interpretation where $(\exists x P(x)) \wedge (\exists x Q(x))$ is true but $\exists x(P(x) \wedge Q(x))$ is false.

13. Over $\mathbb{R}$, negate $\forall x \exists y\,(x \leq y)$.

14. Identify the free variables in $\forall x(P(x, y) \wedge \exists z R(x, z, w))$.

15. Evaluate $\forall x \in \varnothing P(x)$ and $\exists x \in \varnothing P(x)$.

16. Decide whether $\forall x(P(x) \wedge Q(x))$ is equivalent to $(\forall x P(x)) \wedge (\forall x Q(x))$. Explain.

17. Decide whether $\exists x(P(x) \vee Q(x))$ is equivalent to $(\exists x P(x)) \vee (\exists x Q(x))$. Explain.

18. Premises: $\forall x(A(x) \to B(x))$ and $A(c)$. What follows, and which rules justify it?

14 Detailed solutions

1. $\forall x(S(x) \to R(x))$. A nonstudent does not violate the implication.

2. $\exists x(S(x) \wedge \sim R(x))$. The witness must satisfy both conditions.

3. $\forall x \exists y F(x, y)$. The assignment may depend on the student.

4. $\exists y \forall x F(x, y)$. This requires one assignment common to every student; Problem 3 does not.

5. $\exists x \forall y \sim F(x, y)$. In words, some student finished no assignments.

6. True: for any integer x, choose $y = -x \in \mathbb{Z}$.

7. False: if $x = 0$, then $y = 0$, while if $x = 1$, then $y = -1$; one fixed y cannot do both.

8. $\exists x(L(x) \wedge \forall y(L(y) \to y = x))$.

9. $\exists x \exists y(L(x) \wedge L(y) \wedge x \neq y)$.

10. $\forall x \forall y((L(x) \wedge L(y)) \to x = y)$. No; this is true even when nobody is late.

11. Take $D = \{a, b\}$ and $P(a) = 1, P(b) = 0, Q(a) = 0, Q(b) = 1$. The first formula is true; the second is false.

12. Use the same D, P, Q as in Problem 11. Each existential claim has a witness, but no object satisfies both predicates.

13. $\exists x \forall y(x > y)$. Both quantifiers flip and $\sim (x \leq y)$ becomes $x > y$.

14. y and w are free. The quantifiers bind x and z.

15. The universal statement is true vacuously; the existential statement is false because no witness exists.

16. Yes. The left side says every object satisfies both properties, which is exactly the same as saying every object satisfies P and every object satisfies Q.

17. Yes. An object satisfying $P \vee Q$ satisfies at least one predicate; conversely a witness for either existential claim is a witness for $P \vee Q$.

18. $B(c)$. Universal instantiation gives $A(c) \to B(c)$; modus ponens with $A(c)$ gives $B(c)$.